DAVID CRABTREE

Data Analyst | Causal Inference & Survey Research
Chicago, IL | crabtree.david.dc@gmail.com | davidicrabtree.com/data-science

SUMMARY

Quantitative social scientist specializing in causal inference and survey experiments. Six years of R, from data collection to published analysis; four peer-reviewed articles. Led a 10-person team building a 20,000-observation NSF-funded dataset. Building Python and SQL in Georgia Tech's MS in Analytics.

TECHNICAL SKILLS

Languages: R (advanced), Python (intermediate), SQL (basic)

Libraries: R: tidyverse, ggplot2, estimatr; Python: pandas, scikit-learn, statsmodels, matplotlib

Methods: randomized experiments and A/B testing, treatment effects, IV/2SLS, power analysis, regression, machine learning (lasso, tree ensembles), survey design and weighting, data quality assurance

Tools: Git/GitHub, Quarto, R Markdown, Qualtrics, LaTeX

EXPERIENCE

Data Manager, American Bar Foundation — Spring 2024 – Summer 2025

NSF-funded CIGO Project; PIs Tom Ginsburg and Kristina Daugirdas

- Led 10 research assistants coding 1,200+ charters of 600+ intergovernmental organizations.
- Designed double-coding and reconciliation: every observation coded twice, discrepancies resolved.
- Wrote R scripts to merge the team's coding into a 20,000+ observation dataset and flag anomalies.
- Co-authored the dataset's article (*Journal of Empirical Legal Studies*, forthcoming).

Instructional Assistant, University of Chicago (full-time) — Sep 2023 – Sep 2025

- Led discussion sections, graded, and held office hours for 12 sections of 5 political science courses.
- Advised senior honors theses as preceptor for two cohorts.

Research Assistant, University of Chicago — 2023

- Hand-coded term limits in hundreds of constitutions for a sole-authored *Democratization* article.
- Visualized cross-national data on political repression in R for the Transitional Justice Lab.

Research Assistant, East Tennessee State University — 2021 – 2022

- Co-designed and programmed a randomized survey experiment with a national sample of 1,650.
- Ran all analysis in R and first-authored two peer-reviewed articles from the study.

PROJECTS

Machine-Learning Attrition Weights for the GSS Panel, Python — 2026

github.com/davidicrabtree/gss-panel-attrition

- Modeled dropout from the 2016–2020 GSS panel with lasso, random forests, and gradient boosting.
- Cut attrition bias 53% on 15 held-out benchmarks, versus 11% for NORC's released weight.
- Found the best dropout predictor was not the best weight; attitude-based models cut more bias.

Re-examining the Oregon Health Insurance Experiment, Python and R — 2026

github.com/davidicrabtree/oregon-medicaid-experiment

- Reproduced a landmark Medicaid experiment (*NEJM*, 2013) from public data in Python and R.
- Showed its headline null had only 67% power to detect a clinically significant effect.

EDUCATION

Georgia Institute of Technology, MS in Analytics (online) — Expected Dec 2027

University of Chicago, PhD program in Political Science; exams completed — 2025 – present

University of Chicago, MA in Social Sciences — 2023

East Tennessee State University, BA in Political Science; BA in International Affairs — 2022

PUBLICATIONS

Crabtree, David. 2025. "Can Unamendable Term Limits Prevent Presidents-For-Life?" *Democratization*.

Daugirdas, Kristina, Tom Ginsburg, **David Crabtree**, and Morgen Miller. Forthcoming. "Charters of International Organizations: Evidence From a New Dataset." *Journal of Empirical Legal Studies*.

Crabtree, David, and Wesley Wehde. 2023. "Examining Policy Feedback Effects from COVID-19 on Social Welfare Support: Developing an Outcome Distance Dimension." *Policy & Politics*.

Crabtree, David, and Wesley Wehde. 2022. "Who Deserves What and Why During the COVID-19 Pandemic: Applying the CARIN Principles of Deservingness to the American Welfare State." *Social Policy & Administration*.

AWARDS

$22,000 in research funding: Malyi Center Research Grant ($10,000, 2025), Phi Kappa Phi Graduate Fellowship ($8,500, 2022), ETSU Honors Summer Research Fellowship ($3,500, 2021). Merit scholarships: Maroon Scholar, University of Chicago; President's Academic Performance Scholarship, ETSU.